GreenAudioBench: Reproducible Accuracy, Latency and Energy Benchmarking of Frozen Audio Foundation Models for Environmental Sound Classification

Anonymous Submission
Author names and affiliations omitted for double-blind review

***Abstract*—Model selection for audio classification is still reported almost exclusively through accuracy, even though deployed systems, from smart-city acoustic monitoring to always-on IoT and edge sensing, are also constrained by inference latency and energy draw. We present GreenAudioBench, a reproducible benchmark that places five pretrained audio encoders on a common accuracy, latency and energy axis under a single controlled protocol. All encoders stay frozen and are compared through the same fold-aware linear probe on the five official folds of ESC-50, while latency and GPU board energy are measured for every model on one Tesla T4 with an idle baseline subtracted. Accuracy ranking and efficiency ranking do not coincide. MS-CLAP reaches the highest probe accuracy of 97.95 percent while needing 12.34 ms and 0.725 J per clip at batch size one, whereas AST reaches 96.15 percent at 82.73 ms and 5.013 J per clip. BEATs matches the mean accuracy of AST to two decimal places yet is roughly 3.4 times faster and 3.7 times cheaper in dynamic energy. Only PANNs CNN14 and MS-CLAP remain undominated in the accuracy-energy plane. Half precision is not a portable optimisation: it cuts AST latency and energy by about three quarters but produces invalid embeddings for three of the five implementations. Zero-shot transfer of the two CLAP models is also prompt dependent, with a template change worth 3.60 points to one model and minus 1.15 points to the other. We release the measurement protocol, provenance metadata and result files so the numbers can be audited and extended.**

***Index Terms*—audio classification, foundation models, energy efficiency, inference latency, benchmarking, reproducibility, edge deployment, smart cities**

I. Introduction

Environmental sound classification underpins a growing class of always-on services in intelligent digital ecosystems: acoustic monitoring in smart cities, anomaly detection on industrial IoT sites, wildlife sensing and context awareness on consumer devices. These services are increasingly built by reusing a large pretrained audio encoder rather than by training a task-specific network from scratch. Encoders such as PANNs [3], AST [4], BEATs [5] and the contrastive language-audio models LAION-CLAP [6] and MS-CLAP [7] are now the default starting point.

The literature that compares these encoders is, however, organised almost entirely around one number. A model is reported as better because it is one or two points more accurate on a benchmark, and the cost of obtaining those points is left implicit. This is a poor match for how the models are actually deployed. An acoustic sensor that must run continuously on a small accelerator is constrained by how many milliseconds a clip takes and by how many joules that clip costs, and a two-point accuracy gain that multiplies energy per clip by seven is not obviously a gain at all. The wider argument for reporting efficiency alongside quality has been made repeatedly [10], [11], [12], and standardised inference measurement exists for general machine learning workloads [13], but audio representation benchmarks such as HEAR [9] still rank systems on quality alone.

Combining published numbers does not close this gap. Accuracy figures for these encoders come from different fine-tuning recipes, different splits and different augmentation policies, while whatever timing information exists was produced on different accelerators, drivers and framework versions. Nothing in such a table is comparable across rows.

This paper takes the opposite approach and measures everything once, under one protocol. Our contributions are the following.

- **A controlled quality-efficiency protocol.** Five pretrained audio encoders are held frozen and compared through an identical fold-aware linear probe on the five official folds of ESC-50 [1], so that differences reflect the representations rather than differences in downstream training.
- **Measured rather than estimated cost.** Latency and GPU board energy are measured for every model on the same Tesla T4, with a common waveform-to-embedding boundary, a measured idle baseline and both total and idle-subtracted energy per clip.
- **An operating-point analysis.** We compute which models remain undominated in the accuracy-latency and accuracy-energy planes, and show that accuracy ranking and efficiency ranking separate.
- **Two deployment-relevant negative results.** Half precision fails outright for three of the five implementations in our pipeline, and batching helps some encoders while actively hurting others.
- **Auditable provenance.** Dataset checksum, checkpoint revision, commit hash, driver and framework version are written into every result row.

II. RELATED WORK

Large-scale pretraining on AudioSet [2] first showed that audio representations transfer across tasks [19]. The PANNs family [3] established transferable embeddings from log-mel input; AST [4] replaced the convolutional stack with a patch-based spectrogram transformer, later extended with self-supervised objectives [21]; and BEATs [5] added self-supervised pretraining with an acoustic tokenizer. In parallel, the vision-language recipe of CLIP [8] was carried over to audio by MS-CLAP [7] and LAION-CLAP [6], the latter built on a hierarchical audio transformer [20]; both align audio and text in a shared space and therefore also permit classification without any labelled training data.

The standard instrument for comparing frozen representations is the linear probe, which HEAR [9] applied at scale across many audio tasks. Probing isolates representation quality from downstream optimisation, which is exactly the property we need, but HEAR and comparable efforts report quality only. On the cost side, Schwartz et al. [10] argued that efficiency should be a first-class evaluation criterion, Strubell et al. [11] quantified the cost of large-scale NLP training, Henderson et al. [12] proposed systematic energy and carbon reporting, and MLPerf Inference [13] standardised latency and throughput measurement for general workloads. Proxy metrics such as parameter count and multiply-accumulate operations are frequently substituted for measurement, but they ignore memory traffic, kernel efficiency and front-end preprocessing, and Section IV-C shows the substitution is unsafe for this model family; mixed-precision inference [14] is likewise usually assumed to transfer, and Section IV-E shows that it does not.

To our knowledge no prior work reports accuracy, measured latency and measured GPU energy for this set of audio encoders under one frozen protocol on one accelerator.

III. BENCHMARK DESIGN

A. Dataset and folds

All reported results use ESC-50 [1]: 2000 clips of five seconds each, 50 balanced classes, approximately 2.78 hours of audio. ESC-50 ships with five official folds constructed so that clips originating from the same source recording never cross the fold boundary. We use those folds unchanged and never re-split the data, which keeps our numbers comparable with the ESC-50 literature and avoids the optimistic bias of random splitting. The dataset archive is pinned to a fixed source revision and verified by SHA-256 before use. Each outer fold therefore contributes 1600 training clips and 400 test clips.

B. Models and checkpoint identity

Five encoders are evaluated: AST [4], BEATs [5], PANNs CNN14 [3], LAION-CLAP [6] and MS-CLAP [7]. Because published checkpoints for the same architecture can differ substantially, the exact checkpoint revision is recorded in every result row rather than being described only by model name. The BEATs checkpoint required a cross-verified mirror because the original vendor URL was unavailable; this substitution is recorded in the released metadata rather than hidden.

C. Shared decoding, model-specific front-ends

A single deterministic decode path is applied before any model-specific processing: decode, channel-mean downmix to mono, explicit resampling and conversion to a float waveform, with no per-clip peak normalisation. After that point every encoder keeps its own official front-end. AST keeps its reference feature extractor, BEATs keeps its vendored preprocessing including the required waveform scaling, PANNs keeps its log-mel front-end inside the measured path, and both CLAP models keep their expected pipelines.

This split is deliberate. Forcing one generic feature pipeline on all five encoders would silently degrade the models whose weights were trained against a different front-end, and would turn a representation comparison into a preprocessing-mismatch comparison. Keeping the front-ends inside the measured region also means that reported latency and energy are the cost of the deployable unit, not of an idealised tensor-in tensor-out core.

D. Frozen linear probe

No encoder is fine-tuned. Embeddings are extracted once, standardised, and classified by multinomial logistic regression (`lbfgs`, `max_iter` = 2000) from scikit-learn [15]. The regularisation strength is searched over $C \in \{0.01, 0.1, 1, 10, 100\}$.

Hyperparameter selection is fold-aware. For each outer fold, one official fold is held out, the remaining four form the training pool, and C is selected by inner cross-validation restricted to those four folds. The outer test fold never participates in selection, so no test information leaks into the probe.

E. Variability reporting

Because the probe is deterministic under `lbfgs` on fixed embeddings, repeating it with different seeds does not produce meaningful variation, and reporting such repetitions as a variance estimate would be misleading. We therefore report one value per official outer fold, and every interval in this paper is the standard deviation *across the five official folds*. We make no claims of statistical significance, as five folds do not support a meaningful test between closely spaced systems.

F. Weak zero-shot protocol

The two CLAP models additionally support classification without any labelled training data by scoring audio embeddings against text embeddings of class prompts. We evaluate two templates, "`a sound of {class_name}`" (primary) and "`this is the sound of {class_name}`" (alternative), over the same five official folds.

We label this setting *weak zero-shot*. Both CLAP models were pretrained on large audio-text corpora whose contents are not fully disclosed and may overlap with the source recordings behind ESC-50. The condition is therefore zero-shot with respect to our pipeline, but not verifiably zero-shot with respect to pretraining exposure, and the resulting numbers should be read as an upper bound.

G. Latency protocol

Latency is measured from waveform input to embedding output, so every model pays for its own front-end. Each configuration runs a warm-up stage followed by 100 timed iterations repeated three times, giving 300 timed iterations per row, with explicit CUDA synchronisation around each timed region. We report the median per clip together with the 25th and 75th percentiles, since the distribution is not symmetric. Batch sizes 1 and 32 are measured to separate the latency-oriented and throughput-oriented regimes.

H. GPU energy protocol

GPU board power is sampled through NVML [17] at approximately 10 Hz and integrated over time. Every configuration is measured over a window of at least 30 s, yielding between 279 and 301 power samples per row, which avoids the aliasing that short windows produce against a sampler of this rate. Energy is divided by the number of clips actually processed inside the window rather than by a nominal count.

A separate idle baseline of at least 30 s is measured once for the whole run and gives 9.976 W on our device. We report both total energy per clip and *dynamic* energy per clip, the latter with the idle baseline subtracted. The distinction matters: on a device whose idle draw is close to 10 W, a cheap model spends a large fraction of its total figure simply keeping the board powered, so total energy understates the differences between models while dynamic energy isolates the work attributable to inference. GPU temperature is recorded before and after each configuration and stayed between 60 °C and 79 °C, so no configuration was measured under thermal throttling.

Two caveats are structural. NVML reports GPU board power, not whole-system power, so host CPU, memory and PSU losses are excluded. And we deliberately do not convert joules into carbon [22], because that conversion depends on grid intensity and total system draw, neither of which we measure.

I. Provenance

Every result row carries dataset checksum, checkpoint revision, git commit, a clean-working-tree flag, Python 3.12.13, PyTorch [16] 2.11.0+cu128, CUDA 12.8, driver 580.82.07 and the GPU name. All efficiency measurements come from a single Tesla T4. Configurations that failed are kept in the result files with their error status rather than dropped, so that failures are visible in the released evidence.

IV. Results

A. Frozen-probe accuracy

Table I reports probe accuracy and macro-F1. The two CLAP encoders lead: MS-CLAP reaches 97.95 % and LAION-CLAP 97.20 %, both with a fold spread below one point. AST and BEATs follow at 96.15 %, identical to two decimal places, and PANNs CNN14 trails at 92.25 %. Macro-F1 tracks accuracy closely everywhere, which is expected on a class-balanced dataset and indicates that no model achieves its score by neglecting a subset of classes.

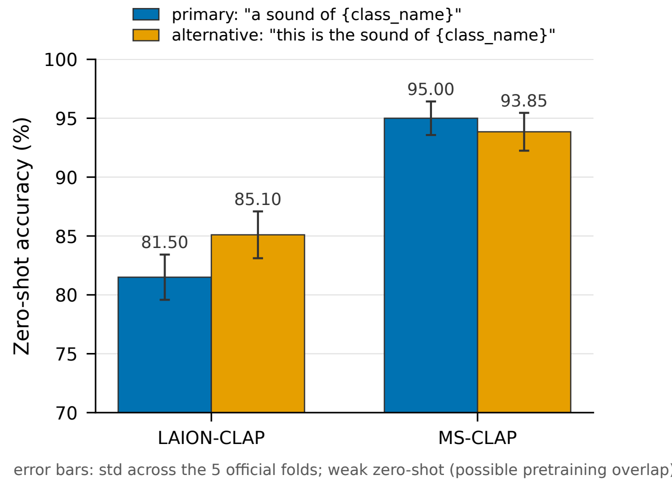

Fig. 1. Weak zero-shot accuracy under the two prompt templates. The same wording change moves the two models in opposite directions.

The AST and BEATs coincidence is the most useful row pair in the table. Two encoders of comparable size produce representations of indistinguishable mean quality under an identical probe, which turns the efficiency comparison between them into an almost controlled experiment: whatever separates them in Section IV-C cannot be explained away as a quality difference. We note that AST also has the widest fold spread at 2.02 points, roughly 2.7 times that of MS-CLAP, so its representation is the least stable across the official folds.

B. Weak zero-shot accuracy and prompt sensitivity

Table II and Fig. 1 give the weak zero-shot results. MS-CLAP reaches 95.00 % without any labelled training data, only 2.95 points below its own linear probe, whereas LAION-CLAP reaches 85.10 % at best and stays 12.10 points below its probe. The two models therefore offer very different propositions: for MS-CLAP the label-free path is a genuine alternative on this dataset, while for LAION-CLAP the probe is clearly worth its supervision cost.

Prompt sensitivity does not generalise across models either. Switching from the primary to the alternative template gains LAION-CLAP 3.60 points but costs MS-CLAP 1.15 points. A prompt tuned on one CLAP implementation is thus not portable to another, and any zero-shot number reported without its template is under-specified. Both figures should still be read against the pretraining-overlap caveat of Section III-F.

C. Accuracy against latency and energy

Figs. 2 and 3 place the five encoders in the accuracy-latency and accuracy-energy planes at fp32 and batch size one. A configuration is dominated when another reaches at least equal accuracy at no greater cost, with at least one strict improvement.

TABLE I

QUALITY AND MEASURED FP32 EFFICIENCY ON ESC-50, TESLA T4. ACCURACY AND MACRO-F1 ARE MEANS OVER THE FIVE OFFICIAL FOLDS WITH THE STANDARD DEVIATION ACROSS FOLDS. LATENCY IS THE MEDIAN PER CLIP OVER 300 TIMED ITERATIONS. ENERGY IS DYNAMIC JOULES PER CLIP, WITH THE 9.976 W IDLE BASELINE SUBTRACTED; TOTAL JOULES PER CLIP ARE GIVEN IN PARENTHESES. BEST VALUE IN EACH COLUMN IS IN BOLD.

					Latency (ms/clip)		Dynamic energy (J/clip)	
Model	Params (M)	GMACs	Accuracy (%)	Macro-F1 (%)	batch 1	batch 32	batch 1	batch 32
MS-CLAP	33.2	7.0	**97.95** ± 0.76	**97.89** ± 0.83	**12.34**	9.44	0.725 (0.856)	0.513 (0.607)
LAION-CLAP	**27.5**	**5.9**	97.20 ± 0.78	97.17 ± 0.79	44.51	36.31	1.461 (1.948)	1.479 (1.860)
AST	86.2	103.5	96.15 ± 2.02	96.06 ± 2.13	82.73	99.78	5.013 (5.898)	5.338 (6.330)
BEATs	90.3	23.6	96.15 ± 1.45	96.08 ± 1.50	24.05	21.61	1.365 (1.615)	1.192 (1.407)
PANNs CNN14	81.8	10.5	92.25 ± 1.33	92.15 ± 1.45	12.67	**4.89**	**0.706** (0.836)	**0.280** (0.330)

TABLE II

WEAK ZERO-SHOT ACCURACY ON ESC-50 (MEAN AND STANDARD DEVIATION ACROSS THE FIVE OFFICIAL FOLDS). SEE SECTION III-F FOR WHY THIS SETTING IS CALLED WEAK.

Model	Template	Accuracy (%)	Macro-F1 (%)
LAION-CLAP	primary	81.50 ± 1.92	79.35 ± 2.10
LAION-CLAP	alternative	85.10 ± 1.99	83.11 ± 2.13
MS-CLAP	primary	95.00 ± 1.43	94.79 ± 1.64
MS-CLAP	alternative	93.85 ± 1.61	93.51 ± 1.89

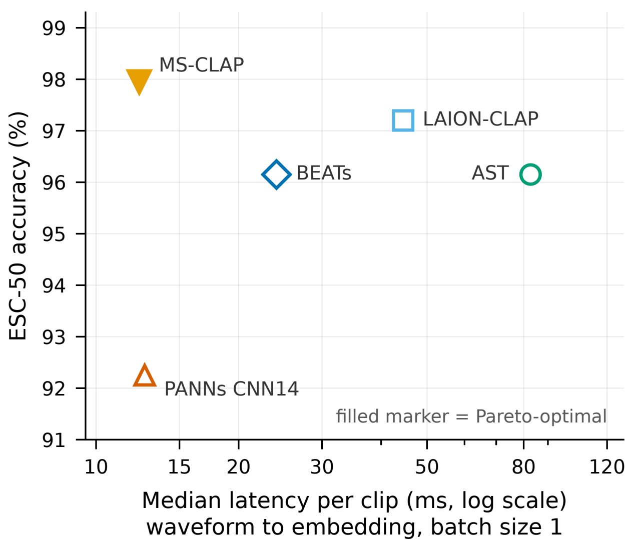

Fig. 2. Accuracy against median latency per clip, fp32, batch size 1. Filled markers are undominated. The horizontal axis is logarithmic.

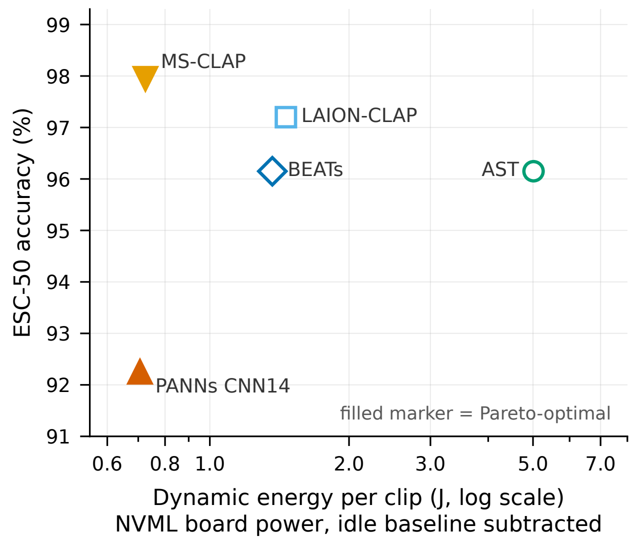

Fig. 3. Accuracy against dynamic energy per clip, fp32, batch size 1, NVML board power with the idle baseline subtracted. Filled markers are undominated. The horizontal axis is logarithmic.

Under that rule only MS-CLAP is undominated in the accuracy-latency plane at batch size one, and MS-CLAP together with PANNs CNN14 is undominated in the accuracy-energy plane. At batch size 32 both frontiers reduce to the same pair. The remaining three encoders are dominated under these conditions.

The energy frontier is also unusually tight, and the tightness is itself a result. At batch size one PANNs CNN14 is cheaper than MS-CLAP by only 2.7 %, 0.706 against 0.725 J per clip, while giving up 5.70 accuracy points for that saving. The two undominated points are thus nearly indistinguishable in cost and far apart in quality, so being on the frontier is not by itself a reason to prefer the cheaper encoder: at this batch size the low-accuracy end is a poor bargain. Batching changes the picture, because PANNs CNN14 responds to it far better than MS-CLAP does. At batch size 32 the same comparison widens to a factor of 1.83, 0.280 against 0.513 J per clip, which is where a 5.70-point concession starts to buy something. Whether the cheap end of the frontier is worth taking therefore depends on the serving regime, not on the frontier membership alone.

Two comparisons carry most of the practical content. First, AST and BEATs have identical mean probe accuracy, yet BEATs runs 3.44 times faster (24.05 against 82.73 ms per clip) and uses 3.67 times less dynamic energy (1.365 against 5.013 J per clip) at batch size one. When quality is tied, the efficiency gap is large enough to make the choice unambiguous. Second, MS-CLAP is simultaneously the most accurate model and 6.7 times faster and 6.9 times cheaper than AST, which shows that the accuracy-efficiency trade-off in this family is not a smooth frontier that must be paid along.

The proxy metrics predict this poorly. BEATs has the largest parameter count in the table, 90.3 M against 86.2 M for

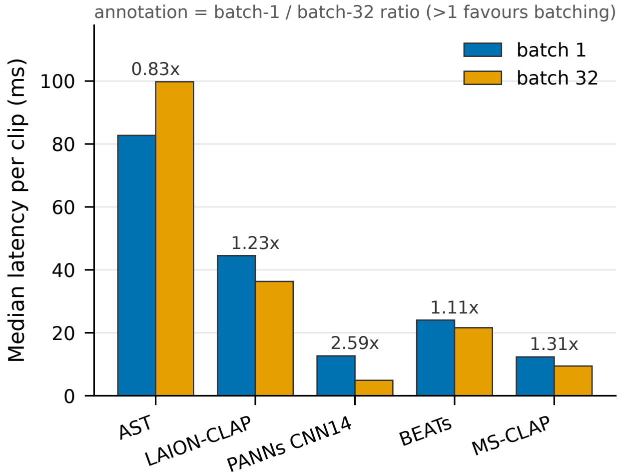

Fig. 4. Effect of batching on median latency per clip at fp32. Annotations give the ratio of the batch-1 value to the batch-32 value, so values above one favour batching. AST is the only model whose latency per clip degrades when batched; dynamic energy follows the same pattern (ratios in the text; per-clip values in Table I).

AST, and is nonetheless the far cheaper of the two, because AST carries 103.5 GMACs against 23.6. Yet GMACs are not reliable either: PANNs CNN14 has 10.5 GMACs against 7.0 for MS-CLAP, a ratio of 1.5, while their measured batch-1 latencies are 12.67 and 12.34 ms, essentially identical. Neither proxy substitutes for measurement here, which is consistent with the fact that model-specific front-ends, memory traffic and kernel efficiency all sit inside the measured region.

D. Effect of batching

Fig. 4 shows what batch size 32 buys. The answer is strongly model dependent. PANNs CNN14 gains most, with latency per clip falling by a factor of 2.59 and dynamic energy by 2.52; MS-CLAP gains 1.31 and 1.41 respectively; BEATs and LAION-CLAP gain little, between 0.99 and 1.15.

AST moves the wrong way. Its latency per clip rises from 82.73 to 99.78 ms when batched, a ratio of 0.83, and its dynamic energy per clip also rises slightly. On a 16 GB T4, batching 32 spectrogram-transformer inputs pushes the model into a regime where the extra memory pressure outweighs the gain in parallel occupancy, so the usual assumption that throughput-oriented serving should batch aggressively does not hold for every encoder on every accelerator. Batch size is a parameter that has to be measured per model rather than set by convention.

E. Half-precision compatibility

Half precision is often treated as a free lever [14]. In our pipeline it is not. Table III reports what happened when each configuration was re-measured at fp16.

Three of the five implementations, BEATs, MS-CLAP and PANNs CNN14, produced NaN or Inf values in their embeddings at both batch sizes and yielded no usable measurement. We keep these configurations in the released result files with their error status rather than omitting them, because a table that silently contains only the successes would misrepresent how portable fp16 is. We attribute the failures to the numerical range of specific operations in these particular implementations under naive casting rather than to the architectures themselves; a more careful mixed-precision treatment might well recover them, and that is precisely the engineering cost the naive lever is assumed to avoid.

TABLE III
HALF-PRECISION INFERENCE. ONLY TWO OF THE FIVE IMPLEMENTATIONS PRODUCED VALID EMBEDDINGS AT FP16. CHANGES ARE RELATIVE TO FP32, SO NEGATIVE VALUES ARE CHEAPER.

Model	Batch	Latency change	Dynamic energy change
AST	1	-75.5 %	-74.7 %
AST	32	-74.8 %	-75.6 %
LAION-CLAP	1	$+1.0$ %	-8.6 %
LAION-CLAP	32	-12.8 %	-35.3 %
BEATs	1, 32	NaN or Inf in embeddings	
MS-CLAP	1, 32	NaN or Inf in embeddings	
PANNs CNN14	1, 32	NaN or Inf in embeddings	

Where fp16 worked, the benefit was still uneven. AST gains the most, shedding 75.5 % of its latency and 74.7 % of its dynamic energy at batch size one, which moves it from the most expensive model in the study to a competitive one. LAION-CLAP gains far less and is batch dependent: at batch size one its latency is unchanged within noise at $+1.0$ %, while dynamic energy falls 8.6 %; at batch size 32 the gains grow to -12.8 % and -35.3 %. Half precision is therefore an implementation-specific and configuration-specific optimisation that must be validated per model, not a portable default.

V. DISCUSSION

The central observation is that the accuracy ranking and the cost ranking of these encoders are close to independent. Ordering the five models by probe accuracy yields MS-CLAP, LAION-CLAP, AST, BEATs, PANNs CNN14; ordering them by dynamic energy at batch size one yields PANNs CNN14, MS-CLAP, BEATs, LAION-CLAP, AST. AST is third by quality and last by cost, by nearly an order of magnitude. Selecting a model from an accuracy table alone therefore carries a cost penalty that the table cannot express.

For practitioners the results map onto three concrete regimes. Where quality dominates and a GPU is available, MS-CLAP is the straightforward choice, since it is both the most accurate model measured and among the cheapest, and it additionally offers a label-free path within three points of its own probe. Where energy is the binding constraint, for instance on battery-powered or thermally-limited always-on sensing, PANNs CNN14 delivers the lowest cost measured, at 0.280 J per clip in exchange for 5.70 accuracy points, but only where the workload can be batched: at batch size one its energy advantage over the far more accurate MS-CLAP collapses to 2.7 %. Where a spectrogram transformer is required for other reasons, BEATs dominates AST outright at fp32 under our conditions, matching its accuracy at roughly a quarter of the cost.

Three methodological points generalise beyond this model set. Proxy metrics are unsafe: neither parameter count nor GMACs ordered our models correctly against measured latency, because the deployable unit includes a model-specific front-end and is shaped by memory traffic that the proxies do not see. Configuration defaults are unsafe: batching helped by a factor of 2.59 for one model and hurt another, and fp16 ranged from a three-quarter reduction to outright failure. And negative results carry information: the three fp16 failures are arguably the most actionable finding here for anyone planning a half-precision deployment, and they exist in this paper only because failed configurations were retained rather than dropped.

VI. Limitations

The scope of the evidence should be read narrowly. Our official numerical evaluation covers ESC-50 only; the accuracy ordering may differ on corpora with different acoustic characteristics, such as UrbanSound8K [18], about which we make no claim. All efficiency numbers come from a single Tesla T4 with one driver and framework stack, and both the ordering and the batching behaviour could change on an accelerator with different memory bandwidth or hardware support. The compared encoders differ in architecture, pretraining corpus and checkpoint design simultaneously, so we describe what the deployable artefacts do without attributing the differences to any single causal factor; frozen probing measures representation quality, not fine-tuning potential. Variability is reported across five official folds and we make no claims of statistical significance. The zero-shot evaluation covers only the two CLAP-family models with two simple templates, and carries the pretraining-overlap caveat of Section III-F. Finally, NVML measures GPU board power rather than whole-system power, dynamic energy per clip depends on the measured idle baseline, and we make no carbon claims.

VII. Conclusion

We presented GreenAudioBench, a controlled benchmark that measures accuracy, latency and GPU energy for five frozen pretrained audio encoders under one protocol on ESC-50 and one Tesla T4. Accuracy and efficiency rankings separate: only PANNs CNN14 and MS-CLAP remain undominated, and two encoders with identical mean accuracy differ by a factor of roughly 3.5 in latency and energy. Common efficiency levers did not transfer, with batching helping one model by 2.59 times while degrading another, and half precision failing outright for three of five implementations. Proxy metrics did not order the models correctly against measurement.

The practical conclusion is that quality-only leaderboards are insufficient for deployment-oriented model selection in this domain, and that efficiency claims must be measured on the target stack rather than assumed. Future work will extend the protocol to a second dataset and to accelerators outside the T4 class, and will examine whether careful mixed-precision treatment recovers the three failing implementations.

References

[1] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. 23rd ACM Int. Conf. Multimedia*, 2015, pp. 1015–1018.

[2] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio Set: An ontology and human-labeled dataset for audio events," in *Proc. IEEE ICASSP*, 2017, pp. 776–780.

[3] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "PANNs: Large-scale pretrained audio neural networks for audio pattern recognition," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 28, pp. 2880–2894, 2020.

[4] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio Spectrogram Transformer," in *Proc. Interspeech*, 2021, pp. 571–575.

[5] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, W. Che, X. Yu, and F. Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. 40th Int. Conf. Machine Learning*, PMLR vol. 202, 2023, pp. 5178–5193.

[6] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE ICASSP*, 2023, pp. 1–5.

[7] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in *Proc. IEEE ICASSP*, 2023, pp. 1–5.

[8] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Machine Learning*, 2021, pp. 8748–8763.

[9] J. Turian, J. Shier, H. R. Khan, B. Raj, B. W. Schuller, C. J. Steinmetz, et al., "HEAR: Holistic evaluation of audio representations," in *Proc. NeurIPS 2021 Competitions and Demonstrations Track*, PMLR vol. 176, 2022, pp. 125–145.

[10] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.

[11] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proc. 57th Annu. Meeting Assoc. Computational Linguistics*, 2019, pp. 3645–3650.

[12] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," *J. Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.

[13] V. J. Reddi, C. Cheng, D. Kanter, P. Mattson, G. Schmuelling, C.-J. Wu, et al., "MLPerf inference benchmark," in *Proc. ACM/IEEE 47th Annu. Int. Symp. Computer Architecture (ISCA)*, 2020, pp. 446–459.

[14] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, "Mixed precision training," in *Proc. Int. Conf. Learning Representations*, 2018.

[15] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., "Scikit-learn: Machine learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[16] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, et al., "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019, pp. 8024–8035.

[17] NVIDIA Corporation, "NVIDIA Management Library (NVML)," 2024. [Online]. Available: https://developer.nvidia.com/nvidia-management-library-nvml

[18] J. Salamon, C. Jacoby, and J. P. Bello, "A dataset and taxonomy for urban sound research," in *Proc. 22nd ACM Int. Conf. Multimedia*, 2014, pp. 1041–1044.

[19] S. Hershey, S. Chaudhuri, D. P. W. Ellis, J. F. Gemmeke, A. Jansen, R. C. Moore, et al., "CNN architectures for large-scale audio classification," in *Proc. IEEE ICASSP*, 2017, pp. 131–135.

[20] K. Chen, X. Du, B. Zhu, Z. Ma, T. Berg-Kirkpatrick, and S. Dubnov, "HTS-AT: A hierarchical token-semantic audio transformer for sound classification and detection," in *Proc. IEEE ICASSP*, 2022, pp. 646–650.

[21] Y. Gong, C.-I. J. Lai, Y.-A. Chung, and J. Glass, "SSAST: Self-supervised audio spectrogram transformer," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 36, no. 10, 2022, pp. 10699–10709.

[22] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," 2019, arXiv:1910.09700.